

JAVA SNACKS – BITTERKOLA

(Monday, April 6, 2026)



Instructions:

1. **Avoid using Artificial Intelligence(AI) tools.**
2. **Submission must occur EVERYDAY.**
3. **Each question is to be in a different file.**
4. **Each file is to be titled what the app is doing e.g QuestionOne can be named TimeFormatter**
5. **For every day create a new folder in your local machine and save each file in the respective folders.**

Submission Guideline: It is safe to assume you have created a repo called JavaTasks(or its equivalent).

Inside our folder called easterEggs create the folder bitterKola and save all of today's tasks inside it

Push to git and paste the link to the repository on the provided Google Sheet document. (Put the link on Column G).

1. Write a program that reads two times in HH:MM format (as separate integers for hours and minutes) and displays the total time after adding them, handling carry-over of minutes into hours.
2. Write a program that prompts the user to enter a 5-digit integer and computes the sum of the first and last digits.
3. Write a program that reads a price and a discount percentage, calculates the discount amount, and displays the final price.
4. Write a program that reads two integers and determines whether the first is a multiple of the second.
5. Write a program that takes a person's weight (in kg) and height (in meters), computes BMI, and classifies as: Underweight (< 18.5), Normal (18.5–24.9), Overweight (25–29.9), Obese (30+).
6. Write a program that reads two integers and prints: the sum if both are positive, the product if both are negative, and the difference (larger minus smaller) if they have opposite signs.
7. Write a program that reads an integer and displays: Buzz if divisible by 5, Fizz if divisible by 3, FizzBuzz if divisible by both, or the number itself if divisible by neither.
8. Write a program that reads a 4-digit PIN entered by the user and validates it: the PIN must be exactly 4 digits (between 1000 and 9999). Display: Valid PIN or Invalid PIN.

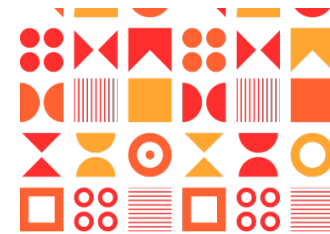
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9. Write a program that prompts the user to enter a date as three integers (day, month, year) and determines whether the date is valid.
10. Write a program that reads an employee's monthly salary and computes the annual tax as follows: no tax on the first ₦300,000; 15% on ₦300,001–₦600,000; 25% on anything above ₦600,000. Display the annual tax owed.
11. Write a program that prompts the user to enter a speed in km/h and determines whether the vehicle is: Stationary (0), Slow (1–40), Moderate (41–80), Fast (81–120), or Dangerously Fast (above 120).
12. Write a program that takes a number of minutes as input and converts it to days, hours, and remaining minutes.
13. Write a program that takes a person's birth year and the current year and determines their age. Then print whether they are eligible for a senior citizen discount (65+).
14. Write a program that accepts a dollar amount within the range 1 to 10(both inclusive) and displays it in words . (e.g., \$5 → five dollars.)
15. Write a program that prompts the user to enter a number and determines whether it is between 10 and 100, exclusive of both endpoints.
16. Write a program that reads two positive integers and determines whether the first is a factor of the second.
17. Write a program that reads three integers and displays them in descending order.
18. Write a program that reads a positive integer and determines whether it is a perfect square (i.e., its square root is a whole number).
19. Write a program that reads two fractions (each as a numerator and denominator) and displays their sum, difference, product, and quotient in decimal form. Handle division by zero.
20. Write a program that reads a letter grade (A, B, C, D, or F) and displays the corresponding GPA value (4.0, 3.0, 2.0, 1.0, or 0.0). Display an error for any other input.
21. Write a program that simulates a vending machine. The user enters a product code (1 = Water ₦100, 2 = Juice ₦200, 3 = Soda ₦150, 4 = Snack ₦300) and the amount tendered. Display the change or a message if the amount is insufficient.
22. Write a program that reads a positive integer n and computes: $(n \times (n + 1)) / 2$. **Let me know your observation. Put it as a Java Doc Comment**
23. Write a program that prompts the user to enter a number of bytes and converts it to kilobytes, megabytes, and gigabytes (using 1 KB = 1024 bytes, 1 MB = 1024 KB, 1 GB = 1024 MB).
24. Write a program that reads three exam scores, calculates the weighted average (40% for the highest, 35% for the middle, 25% for the lowest), and displays the grade.



25. Write a program that reads a person's salary and determines the social insurance deduction: 0% for salary $\leq$ ₦50,000; 5% for ₦50,001–₦150,000; 7.5% for ₦150,001–₦500,000; 10% for above ₦500,000.

26. Write a program that reads a positive two-digit integer and determines whether reversing its digits produces a larger number, smaller number, or the same number.

27. Write a program that prompts the user to enter a bank balance and an annual interest rate, then displays the balance after 1, 2, and 3 years of compound interest.

28. Write a program that prompts the user to enter a 3-digit integer and determines whether the number is divisible by the sum of its digits.

29. Write a program that reads two integers and determines which is closer to zero. If they are equally close, print: Both are equally close to zero.

30. Write a program that prompts the user to enter a number of days from today and displays the day of the week it will be. (Assume today is Monday = day 0.)

Have fun guys!